

Unit –I Data Mining

- Define data mining.**
Data Mining is the exploration and analysis of large data sets, in order to discover meaningful patterns and rules.
- What is knowledge discovery in databases?**
Knowledge Discovery in Databases is the process of identifying a valid, potentially useful and ultimately understandable structure in data
- List any four stages of KDD.**
Selection, Preprocessing, Transformation, Data mining, Interpretation and Evaluation, Data Visualization
- What is Loosely Coupled DBMS and Tightly Coupled DBMS?**
 - Loosely-coupled DBMS:** - In this case DBMS is used only for storage and retrieval of data. For example, one can use a loosely-coupled SQL to fetch data records as required by the mining algorithm. The front-end of the application is implanted in a host programming language, with embedded in SQL statements. A loop in the application program copies records in the result set one-by-one from the database address space to the application address space, where computation is performed on them.
 - In tightly-coupled approach,** the portions of the application programs are selectively pushed to the database system to perform the necessary computation. Data are stored in the database and all processing is done at the database end.
- What is prediction and description in Data Mining?**
 - Prediction:** -Predication makes use of existing variables in the database to predict unknown or future values of interest.
 - Description:** -Description focuses on finding patterns describing the data and the subsequent presentation for user interpretation.
- List any four discovery driven tasks.**
Discovery of association rules, Discovery of classification rules, Clustering, Discovery of frequent episode, Deviation detection
- Define Support and Confidence.**
Support: - Support means how often X and Y occur together as a percentage of the total transactions. Let $L = \{I_1, I_2, \dots, I_m\}$ be a set of literal called items. Let D, the database, is a set of transactions, where each transactions T is a set of items. T supports an item x, if x is in T. T is said to support a subset of items X, if T supports each item x in X.

$$\text{Support}(A \Rightarrow B) = P(A \cup B)$$
- Confidence:** - $X \Rightarrow Y$ holds with confidence c, if c% of the transactions in D that support X also support Y.

$$\text{Confidence}(A \Rightarrow B) = P(B|A)$$
- Confidence indicates the number of times the if-then statement is found to be true
- Write the objectives of clustering.**
Objectives of clustering are
 - To uncover natural groupings
 - To initiate hypothesis about the data
 - To find consistent and valid organizations of the data
- Define Rough Set.**
Rough set provides a formal framework for the automated transformation of data into knowledge. The rough set is pair of approximations i.e. lower approximation and upper approximations. The lower approximation is said to be positive cases and the upper approximation is said to be the possible cases. Using these approximations, the theory develops tools to discover rules from the given databases.
- What is sequence mining and spatial data mining?**
Sequence mining is concerned with mining sequence data. It may be noted that in the discovery of association rules, we are interested in finding associations between items irrespective of their occurrence.
Spatial data mining is regarded as a special type of data mining that seeks to perform similar generic functions as conventional data mining tools, but modified to take into account the special features of spatial information.
- Write the subtasks of web mining.**
 - Resource finding:** Retrieving documents intended for the web
 - Information selection and preprocessing:** automatically selecting and preprocessing specific information from resources retrieved from the web
 - Generalization:** to automatically discover general patterns at individual web sites as well as across multiple sites
 - Analysis:** Validation and /or interpretation of the mined patterns

12. Expand KDT and IE.
KDT → Knowledge Discovery in Text
IE → Information Extraction
13. What is web mining and text mining?
Web mining is the use of data mining techniques to automatically discover and extract information from web documents and services.
Text mining or KDT (Knowledge Discovery in Text) was first proposed by Feldman and Dagan in 1996. They suggest that the text documents be structured by means of information extraction, text categorization.
14. List any two issues and challenges in data mining.
Limited information, Noise or missing data, User interaction and prior knowledge, Uncertainty
Size, updates and irrelevant fields
15. List the two application areas of data mining.
 - Business and E-Commerce Data
 - Scientific, Engineering and Health Care Data
16. Explain the use of Data Visualization?
Data Visualization: This makes it possible for the analyst to gain a deeper understandability of the data. It helps users to examine large volumes of data and detect the patterns visually. Visual displays of data such as maps, charts and other graphics representations allow data to be presented compactly to the users.
17. Define Machine Learning?
Machine learning is a method of data analysis that automates analytical model building. It is branch of AI based on the idea that system can learn from data, identify patterns and make decisions with minimal human intervention.
18. Define Supervised and Unsupervised Learning?
 - Supervised learning means learning from examples, where a training set is given which acts as examples for the classes. The system finds a description of each class. Once a description has been formulated, it is used to predict the class of previously unseen objects.
 - Unsupervised Learning is learning from observations and discovery. In this mode of learning, there is no training set or prior knowledge of the classes. The system analyzes the given set of data to observe similarities emerging out of the subsets of the data
19. Explain the two fundamental goals of Data Mining?
Two goals of datamining: Prediction and Description
Prediction: -Predication makes use of existing variables in the database to predict unknown or future values of interest.
Description: -Description focuses on finding patterns describing the data and the subsequent presentation for user interpretation.
20. Define Discovery Model? Give an example?
The discovery model is the system automatically discovering important information hidden in the data. The data is shifted in search of frequently occurring patterns, trends and generalization about the data without intervention or guidance from the user.
Example: a supermarket database, which is mined to discover the particular groups of customers to target for a mailing campaign
21. What is Frequent Episodes? List its types?
Frequent episodes are the sequence of events that occur frequently, close to each other and are extracted from the time sequences.
Types: - Serial episode, parallel episode, non-serial and non-parallel episode
22. What is Deviation Detection? Give example?
Deviation detection is to identify outlying points in a particular data set and explain whether they are due to noise or other impurities being present in the data or due to trivial reasons.
Example: -It is applied in forecasting, fraud detection, customer retention etc.

Answer questions:-

1. Write the difference between KDD and Data Mining. (4)

KDD	Data Mining
Knowledge Discovery in Databases is the process of identifying a valid, potentially useful and ultimately understandable structure in data. It consists of various steps like data selection, cleaning, integration, transformation and data mining. It is iterative.	Data Mining is the exploration and analysis of large data sets, in order to discover meaningful patterns and rules. It is step inside the KDD process which deals with identifying patterns. It is non-iterative.
KDD is the representation of structure of data with which knowledge can be acquired.	It works with large amount of data and is concerned with algorithmic means by which patterns are recognized.

2. Define KDD process. Explain the different stages of KDD. (6)

Knowledge Discovery in Databases is the process of identifying a valid, potentially useful and ultimately understandable structure in data.

The stages of KDD are given below:-

- **Selection:** This stage is concerned with selecting or segmenting the data that are relevant to some criteria. For example, for credit card customer profiling, we extract the type of transactions for each type of customers and we are not interested in details of shop where transaction takes place.
- **Preprocessing:** It is the data cleaning stage where unnecessary information is removed. For example, it is unnecessary to note the sex of a customer when any purchase. When data is drawn from several sources, it is possible that the same information represented in different sources in different formats. This stage reconfigures the data to ensure a consistent format, as there is possibility of inconsistent formats.
- **Transformation:** The data is not merely transferred across, but transformed in order to be suitable for the task of data mining. In this stage, the data is made usable.
- **Data mining:** This stage is concerned with the extraction of patterns from the data.
- **Interpretation and Evaluation:** The patterns obtained in the data mining stage are converted into knowledge, which in turn, is used to support decision-making.

Data Visualization: This makes it possible for the analyst to gain a deeper understandability of the data. It helps users to examine large volumes of data and detect the patterns visually. Visual displays of data such as maps, charts and other graphics representations allow data to be presented compactly to the users.

3. Distinguish between DBMS and Data Mining. (4)

DBMS	Datamining
DBMS supports query languages which are useful for query-triggered data exploration.	data mining supports automatic data exploration
If we know what information we are seeking, a DBMS query is preferable	whereas if we unclearly know the possible patterns, then data mining techniques are useful.
Extraction of detailed and summary data	Knowledge discovery of hidden patterns and insights
DBMS support some primitive data mining tasks	DM is hypothesis testing, where we formulate a hypothesis and test it by shifting through the database. This task can be handled by a DBMS query.

4. Explain interaction of Data Mining Systems with Relational DBMS (4)

There are three different ways in which data mining use a relational DBMS.

- They may not use it all
- They may be loosely coupled
- They may be tightly coupled.

1. A majority of DM systems do not use any DBMS and have their own memory and storage management. They treat the database simply as a data repository from which data is expected to be downloaded into their own memory structures, before the data mining algorithm starts. The advantage is that one can optimize the memory management specific to the data mining algorithm.

2. The second approach is to have loosely-coupled DBMS. In this case DBMS is used only for storage and retrieval of data. For example, one can use a loosely-coupled SQL to fetch data records as required by the mining algorithm. The front-end of the application is implanted in a host programming language, with embedded in SQL statements. A loop in the application program copies records in the result set one-by-one from the database address space to the application address space, where computation is performed on them.

3. In tightly-coupled approach, the portions of the application programs are selectively pushed to the database system, to perform the necessary computation. Data are stored in the database and all processing is done at the database end. It is different from bringing the data from the database to the data mining area. This avoids performance degradation and takes full advantage of database technology. The performance of this approach depends on the way to optimize the data mining process while mapping it to query.

5. Explain the following Data Mining techniques. i) Verification Model ii) Discovery Model (4/6)

- a) Verification model: - In this process of data mining, the user makes a hypothesis and tests the hypothesis on the data to verify its validity. The emphasis is on the user who is responsible for formulating the hypothesis and issuing the query on the data to affirm or negate the hypothesis.
Example: - In a supermarket to identify the section of the population most likely to buy a particular product, the user formulates a hypothesis to identify potential customers and their common characteristics. Historical data about transactions and demographic information can then be queried to reveal comparable purchases and the characteristics shared by those purchasers. The whole operation can be repeated by successive refinement of hypothesis until the required limit is reached.
- b) Discovery Model: -The discovery model is the system automatically discovering important information hidden in the data. The data is shifted in search of frequently occurring patterns, trends and generalization about the data without intervention or guidance from the user. The typical discovery tasks are Discovery of association rules, Discovery of classification rules, Clustering, Discovery of frequent episode, Deviation detection

An example of this model is a supermarket database, which is mined to discover the particular groups of customers to target for a mailing campaign.

6. Write a note on i) Discovery of Association Rules ii) Discovery of Classification Rules (4/6)

- i. Discovery of association rules: -An association rule is an expression of the form $X \Rightarrow Y$, where X and Y are the sets of items. The meaning of such a rule is that the transaction of the database which contains X tends to contain Y. Given a database, the goal is to discover all the rules that have support and confidence greater than or equal to the minimum support and confidence.

Let $L = \{l_1, l_2, \dots, l_m\}$ be a set of literal called items. Let D, the database, is a set of transactions, where each transactions T is a set of items. T supports an item x, if x is in T. T is said to support a subset of items X, if T supports each item x in X. $X \Rightarrow Y$ holds with confidence c, if c% of the transactions in D that support X also support Y. The rule $X \Rightarrow Y$ has supports s in the transaction set D is s% of the transactions in D support $X \cup Y$. Support means how often X and Y occur together as a percentage of the total transactions. Thus only the patterns with a combination of intermediate values of confidence and support provide the user with interesting and previously unknown information.

- ii. Discovery of Classification Rules: -Classification involves finding rules that partition the data into disjoint groups. The input for the classification is the training data set, whose class labels are already known. Classification analyses the training data set and constructs a model based on the class label and aims to assign a class label to the future unlabelled records. Since the class field is known, this type of classification is known as supervised learning. A set of classification rules are generated by such a classification process, which can be used to classify future data and develop a better understanding of each class in the database. We can term this as supervised learning.

There are several classification discovery models. They are the decision trees, neural networks, genetic algorithms and the statistical models like linear/ geometric discriminates. The application includes the credit card analysis, banking.

7. Write a note on i) Clustering ii) Frequent Episodes (4/6)

Clustering: - Clustering is a method of grouping data into different groups, so that the data in each group share similar trends and patterns. Clustering constitutes a major class of data mining algorithms. The algorithm attempts to automatically partition the data space into a set of regions or clusters. The goal of the process is to identify all sets of similar examples of data in some optimal fashion.

The objectives of clustering are To uncover natural groupings, To initiate hypothesis about the data, To find consistent and valid organizations of the data

Example: -A retailer may want to know where the similarities exist in his customer base, so that he can create and understand different groups.

- i. Frequent Episode: -Frequent episodes are the sequence of events that occur frequently, close to each other and are extracted from the time sequences. Given a set R of event types, an event is a pair (A, t) where $A \in R$ is an event type and t is an integer, we can calculate the occurrence time of the event. An event sequence s of R is triple (T_s, T_e, S) , where $T_s < T_e$ are integers. T_s is the starting time and T_e is the ending time.

$S = \{(A_1, t_1), (A_2, t_2), \dots, (A_n, t_n)\}$ is the ordered sequence of events, such that $A_i \in R$ and $T_i \leq t_i \leq T_i$ for all $i = 1, 2, \dots, n-1$. The episodes can be of three types.

- The serial episodes which occur in sequence.
- The parallel episodes in which there are no constraints on the order of the event types A and B given.
- The non-serial and non-parallel episodes which occur in a sequences if the occurrences of A and B precede an occurrence of C, and there is no constraint on the relative order of A and B given.

8. Write a note on i) Genetic Algorithms ii) Rough Set Techniques (4/6)

- Genetic Algorithm:** -Genetic Algorithms was inspired by Darwins theory of evolution. A population of individuals, each representing possible solutions to a problem, is initially created at random. Then pairs of individuals combine (cross over) to produce the offspring for the next generation. A mutation process is also used to randomly modify the genetic structure of some members of each new generation. The algorithm runs to generate solutions for successive generation. The probability of an individual reproducing is proportional to the goodness of the solution it represents. Hence the quality of the solutions in successive generation improves. This method can also be applied to data mining problems. Here the quantity to be minimized is often the number of classification errors on a training set. However, the mining of large data sets by genetic algorithm has only recently become practical due to the availability of high-speed computers.
- Rough Sets techniques:** -This theory is introduced by Pawlak in early 1980s, provides a formal framework for the automated transformation of data into knowledge. The rough set is pair of approximations i.e. lower approximation and upper approximations. The lower approximation is said to be positive cases and the upper approximation is said to be the possible cases. Using these approximations, the theory develops tools to discover rules from the given databases. A wide range of applications use this theory. Medical data analysis, aircraft pilot performance evaluation, image processing and voice recognition are some examples.

9. Write a note on Support Vector Machines (SVM) (4)

Support vector machines (SVM): -SVM is based on statistical learning theory and is increasingly becoming useful in data mining. The main idea is to non-linearly map the data set into a high-dimensional feature space and use a linear discriminator to classify the data. Its success has been demonstrated in the areas of classification and decision-tree construction. Unlike other classification techniques, which attempt to minimize the error of classification, SVMs incorporate structure risk minimization which minimizes an upper bound on the generalized errors. e.g. when two set A and B are linearly separable. The idea is to determine from an infinite number of planes correctly separating A and B, the one which will have the smallest generalization error. SVMs select the plane which maximizes the margin separating the two classes.

10. Explain any two discovery driven tasks. (4/6)

Refer q.no 6 and 7 [any two]

11. Write a note on the following: i) Sequence Mining ii) Web Mining iii) Spatial Data Mining (6)

Sequence mining: - Sequence mining is concerned with mining sequence data. It may be noted that in the discovery of association rules, we are interested in finding associations between items irrespective of their occurrence.

Example: -we may be interested in the association between the purchase of particular brand of soft drinks and the occurrence of stomach upsets. But it is more relevant to identify whether there is some pattern in the stomach upsets which occurs after the purchase of the soft drink. Then one is inclined to infer that the soft drinks causes' stomach upsets. On the other hand, if it is more likely that the purchase of the soft drink follows the occurrence of the stomach upset, then it is probable that the soft drink provides some sort of relief to the user. Thus, the discovery of temporal sequences of events concerns casual relationships among the events in a sequence.

One characteristic of sequence patterns discovery in comparison with trend discovery is the lack of shapes, since the casual impact of a series of events cannot be shaped.

Web Mining: - Web mining is the use of data mining techniques to automatically discover and extract information from web documents and services. This area of research is s so huge today partly due to the interests of various research communities, the tremendous growth of information sources available on the web and the recent interest in e-commerce.

Web mining can be broken down into following subtasks:

- Resource finding:** Retrieving documents intended for the web
- Information selection and preprocessing:** automatically selecting and preprocessing specific information from resources retrieved from the web
- Generalization:** to automatically discover general patterns at individual web sites as well as across multiple sites
- Analysis:** Validation and /or interpretation of the mined patterns

Spatial Data mining: -Spatial data mining is the branch of data mining that deals with spatial (location) data. Spatial data mining is regarded as a special type of data mining that seeks to perform similar generic functions as conventional data mining tools, but modified to take into account the special features of spatial information.

For example we may wish to discover some association among patterns of residential colonies and topographical features. A typical spatial association may look like: "The residential Land pockets are dense in a plain region and rocky areas are thinly populated."

12. List and explain the issues and challenges in Data Mining. (4/6)

The issues and challenges in Data Mining:

- Limited information
- Noise or missing data
- User interaction and prior knowledge
- Uncertainty
- Size, updates and irrelevant fields

Limited information: - A database is often designed for purposes other than that of data mining and sometimes, some attributes which are essential for knowledge discovery of the application domain are not present in the data. Thus, it may be very difficult to discover significant knowledge about a given domain.

Noise and Missing data: - Attributes that rely on subjective or measurement judgments can give rise to errors, such that some examples may be misclassified. Missing data can be treated in a number of ways - simply by disregarding missing values, omitting corresponding records, inferring missing values from known values and treating missing data as a special value to be included additionally in the attribute domain. The data should be cleaned so that it is free of errors and missing data.

User interaction and prior knowledge: - An analyst is a person making use of data by means of the available KDD techniques. Since the KDD process is by definition interactive and iterative, it is challenging to provide a high performance, rapid response environment that also assists the users in proper selection and matching of the appropriate techniques to achieve the goals. There needs to be more human-computer interaction and less emphasis on total automation, which supports expert users. The use of domain knowledge is important.

Uncertainty: - This refers to the severity of error and the degree of noise in the data. Data precision is an important consideration in a discovery system.

Size, update and irrelevant fields: - Databases tend to be large and dynamic, in that their contents are kept changing as information is added, modified or removed. The problem with this, from the perspective of data mining, is how to ensure that the rules are up-to-date and consistent with the most current information.

13. Explain the different application areas of Data Mining. (6)

• **Business and E-Commerce Data:** - This is a major source of category for data mining applications back office, front office and network applications produce large amounts of data about business processes. Using this data for effective decision making remains a fundamental challenge.

Business transactions: - Modern business processes are consolidating with millions of customers and billions of their transactions. Business enterprises require necessary information for their effective functioning in the competitive world. Example:- they would like to know, "Is this transaction fraudulent?" "Which customer is likely to migrate?" and "What product is this customer most likely to buy next?"

Electronic Commerce: - Not only does electronic commerce produce large data sets in which the analysis of marketing patterns and risk patterns is critical, but it also important to do this near real time, in order to meet the demands of online transactions.

• **Scientific, Engineering and Health Care Data:** - Scientific data and metadata tend to be more complex in structure than business data. Scientists and engineers are making increasing use of simulation and systems with application domain knowledge.

Genomic data: - Genomic sequencing and mapping efforts have produced a number of databases which are accessible on the web. There are also a wide variety of other inline databases. Finding relationships between these data sources is another fundamental challenge for data mining.

Sensor data: - Remote sensing satellites and variety of other sensors produce large amounts of geo-referenced data. A fundamental challenge is to understand the relationships, including casual relationships among the data.

Simulation Data: - Simulation is accepted as an important mode of science, supplementing theory and experiment. Simulation also produces huge data sets. Data mining and data intensive computing is proving to be a critical link between theory, simulation and experiment.

Health care data: - Hospital, health care organization, insurance companies accumulate large collections of data about patients and health care related details. Understanding relationships in this data is critical for a wide variety of problems ranging from determining what procedures and clinical protocols are most effective; to how best deliver health care to the maximum number of people.

Web data: - The data on the web is growing not only in volume but also in complexity. Web data now includes not only text, audio, and video material, but also streaming data and numerical data.

Multi-Media Documents: - Today technology for retrieving multimedia items on the web is far from satisfactory. On the other hand, an increasingly large number of matters are on the web and the number of users is also growing. It is becoming harder to extract meaningful information from the archives of multimedia data as the volume grows.

Data Web: - The Web is primarily oriented toward document and their multimedia extensions. HTML is a powerful language for supporting this. The potential exist for the web to prove important for working with data. The Extensible Markup Language (XML) is an emerging language for working with data in networked environment.